Expanded Segmentation Method Comparison and Proxy Benchmarking

Purpose and Scope

This v2 package expands the segmentation-method comparison for the App80, App65, and App81 organoid image datasets. It is saved separately from the first comparison package so the earlier output remains unchanged. The expanded package contains more source-matched examples, text-free figure grids, quantitative proxy benchmarks, a heatmap-style summary figure, and source manifests.

The benchmarking is not a manual-ground-truth accuracy study. Instead, it is a method-routing and quality-control comparison against the final AAPS-selected mask for each source image. This framing is important because the final masks were selected after method routing, visual QC, and human-in-the-loop correction; they represent the selected analysis reference for metric extraction, not an independent annotation standard.

Benchmark Definitions

For each non-raw panel, the corresponding instance mask was compared with the final AAPS-selected mask for the same source image. If mask grids differed, the candidate mask was resized to the final mask grid using nearest-neighbour interpolation before binary overlap was computed. The proxy metrics are:

- Dice to final: binary Dice overlap between the candidate mask and the final selected mask.
- Area ratio to final: total candidate mask area divided by final mask area.
- Instance-count delta: candidate instance count minus final instance count.
- Small-fragment burden: fraction of mask area contained in very small connected components.
- Edge-touch burden: fraction of mask area in components touching the image border.
- Dominant-object fraction: fraction of mask area contained in the largest instance.

Visual QC Assessment from Image Review

The expanded figures were reviewed directly using the Codex image viewer. The main observations are consistent across the visual grids and the proxy metrics.

App80 Y-27632. The early prompted-watershed and CPSAM/SAM-style columns often capture only small local regions or partial edges. In later 10 uM images, they miss large organoid-scale structures almost completely. The final multiscale large-recovery branch recovers the biologically relevant large objects, including broad fused structures. This is the correct direction for growth and recovery metrics, but it also means the final branch should be described as a deliberate large-object recovery strategy rather than a universal manual-like contour.

App65 alginate. The old multiscale branch frequently expands into broad irregular masses around dense brightfield structures. The threshold-hinted Cellpose branch gives cleaner separated candidates in

some images, but the reviewed grid shows strong edge-touching and broad peripheral masks in several fields. The final direct-brightfield branch is more conservative and instance-like, which is better aligned with perimeter, wall-darkness, and peripheral/central measurements.

App81 density. This dataset shows why AAPS method routing is necessary. Threshold-hinted and resized multiscale Cellpose can both mistake large pale droplet boundaries for organoids, especially in low-density panels, or miss true cystic organoids when the size prior is wrong. Image-generation-assisted masks are semantically useful for recognizing cystic structures, but reviewed examples also show stylized masks, merged large objects, and non-final mask geometry. The final stitched hybrid output is the most controlled representation because it uses panel-level method selection and human-in-the-loop replacement only where a candidate method improves the selected mask.

Interpretation for the Manuscript

The expanded comparison supports the AAPS claim that robust organoid quantification requires method routing rather than one fixed segmentation algorithm. App80 required large-object recovery because the biological endpoint is growth and recovery under Y-27632. App65 required a more conservative direct-brightfield branch because perimeter and wall-darkness metrics are sensitive to edge-touching and broad masks. App81 required a stitched hybrid strategy because cystic organoid identity, droplet-boundary exclusion, and instance separation varied substantially across density panels.

The quantitative proxy results provide a compact way to report these routing decisions. For App80, early watershed and CPSAM masks show very low overlap and area relative to the final selected masks, reflecting severe under-coverage of large structures. For App65, old and threshold-hinted branches have moderate Dice-to-final values but substantially higher area ratios or edge-touch burdens, matching the visual over-expansion seen in the grids. For App81, image-generation-assisted masks align better with the final selected masks than the tested Cellpose variants on these examples, but the visual review shows why image generation was not used unguarded: it still requires cleanup, splitting, and panel-level selection.

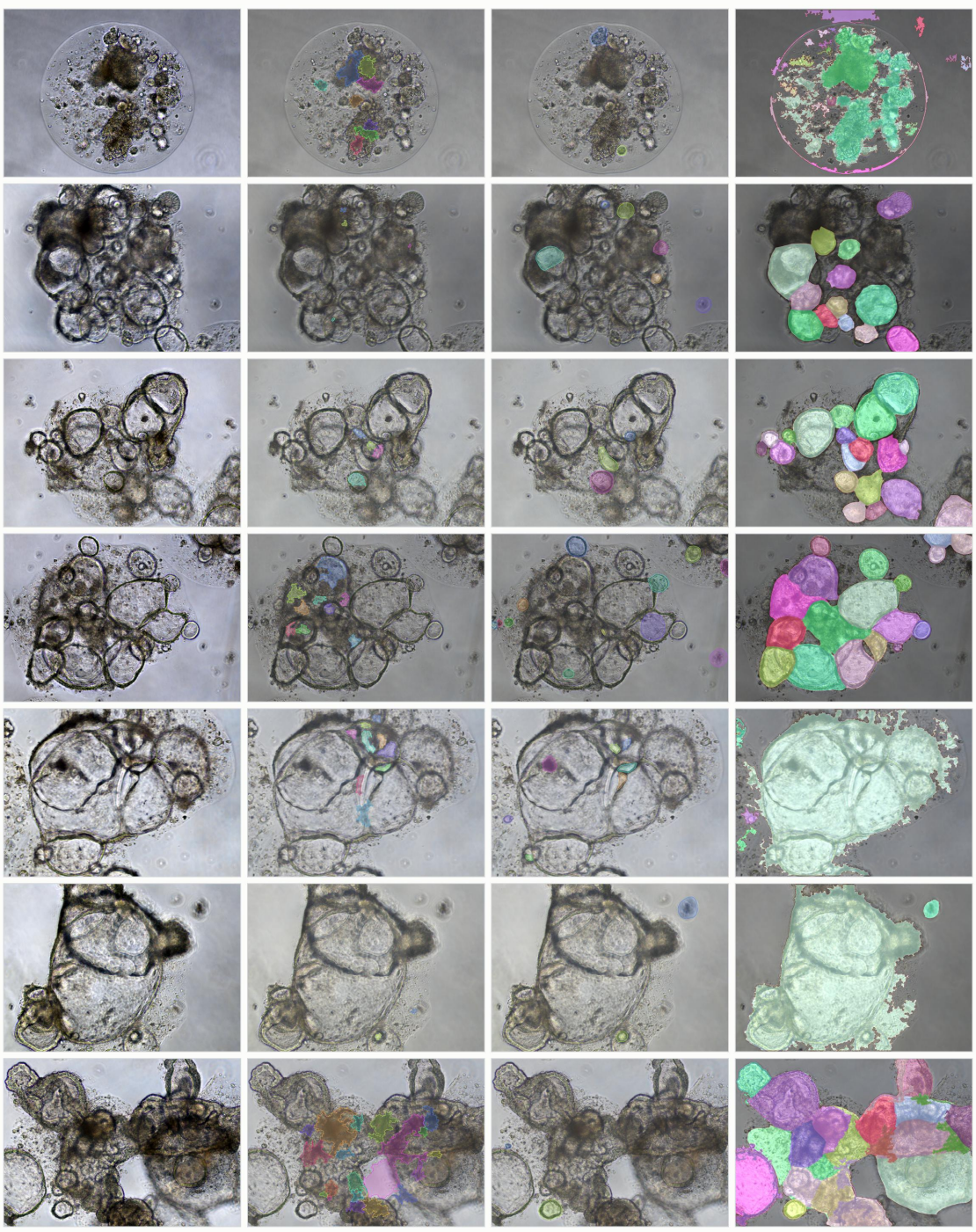


Figure 1: **Expanded App80 Y-27632 segmentation comparison.** Rows show seven source-matched 10 uM Y-27632 fields from 5, 7, 8, 9, 10, 11, and 12 December 2025. Columns show, from left to right: raw brightfield image, early prompted-watershed overlay, early CPSAM/SAM-style proposal overlay, and the final AAPS multiscale large-object recovery overlay. The no-text grid shows that early methods tend to under-cover the large organoid structures that emerge during recovery, while the final branch intentionally recovers larger connected organoid-scale objects for downstream growth and fusion-related metrics.

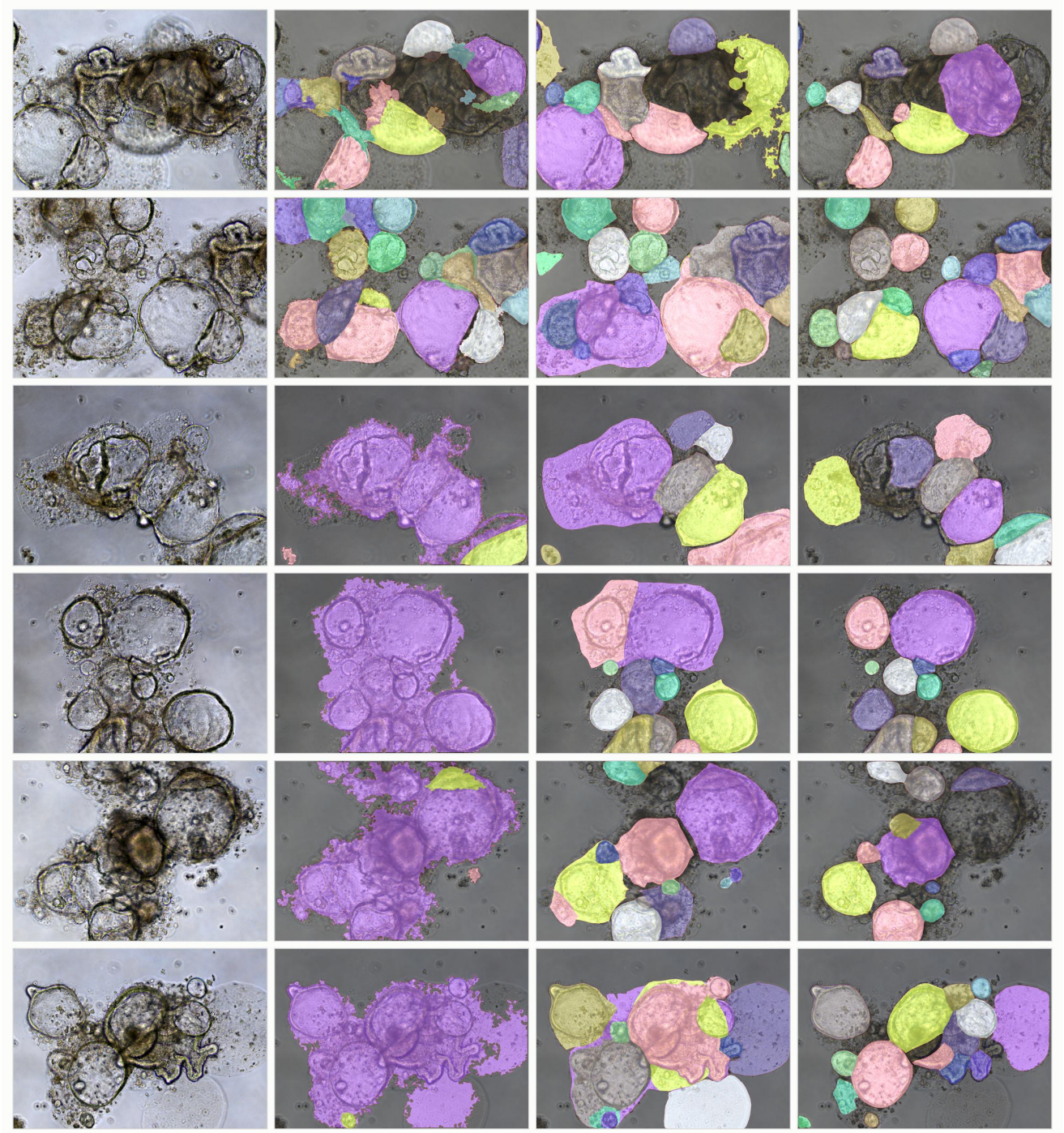


Figure 2: **Expanded App65 alginate segmentation comparison.** Rows show six 0.02% alginate fields from 29 November 2025: 10x00, 10x01, 10x14, 10x15, 10x16, and 10x17. Columns show, from left to right: raw brightfield image, old multiscale Cellpose recovery overlay, threshold-hinted Cellpose overlay, and final AAPS direct-brightfield overlay. The comparison shows that the final branch is more conservative and instance-like than the old large-recovery branch, while avoiding much of the edge-touching burden seen in the threshold-hinted results.

Table 1: Proxy benchmarking against the final AAPS-selected mask. Dice and area ratio are computed after nearest-neighbour resizing to the final mask grid when required. These values are QC/proxy metrics, not manual ground-truth accuracy.

Dataset	Method	n	Dice	Area ratio	Δ count	Small frag.	Edge touch
App80	Prompted watershed	7	0.11	0.03	-9	0.0%	0.0%
App80	CPSAM	7	0.07	0.03	-7	0.0%	0.0%
App80	Final multiscale	7	1.00	1.00	+0	0.0%	0.0%
App65	Old multiscale	6	0.75	1.26	-6	0.0%	34.4%
App65	Threshold-hinted Cellpose	6	0.75	1.49	-0	0.0%	94.8%
App65	Final direct brightfield	6	1.00	1.00	+0	0.0%	9.9%
App81	Threshold-hinted Cellpose	5	0.36	0.67	-2	0.0%	0.0%
App81	Resized multiscale Cellpose	5	0.26	0.92	+0	0.0%	0.0%
App81	Image-generation mask	5	0.48	0.89	+1	2.1%	0.0%
App81	Final stitched hybrid	5	1.00	1.00	+0	0.0%	0.0%

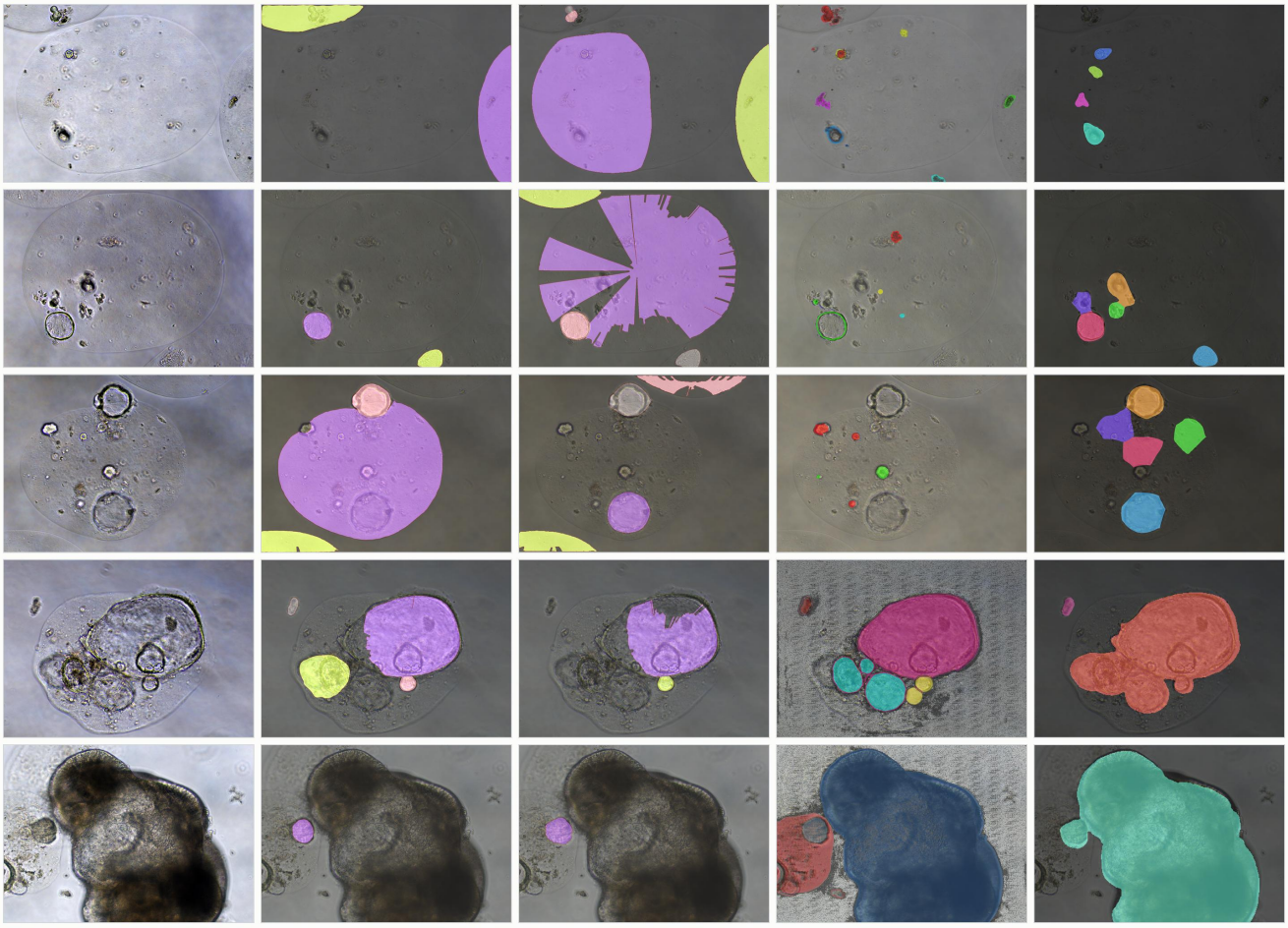


Figure 3: **Expanded App81 density method-routing comparison.** Rows show five source-matched App81 examples: low-density fields from 14, 15, and 16 October 2025, followed by middle-density fields from 18 and 29 October 2025. Columns show, from left to right: raw brightfield image, threshold-hinted Cellpose, resized multiscale Cellpose, image-generation-assisted mask, and final stitched AAPS-selected hybrid overlay. The grid shows that no single method is stable across density, droplet-boundary ambiguity, cystic size range, and large dark organoids. Image generation adds semantic recognition, but the final analysis mask still requires AAPS-style QC and method selection.

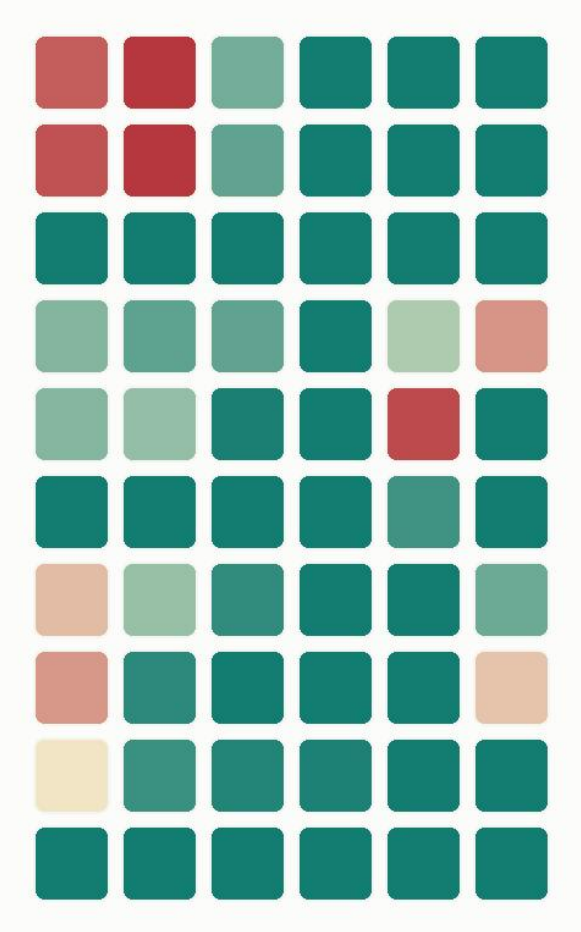


Figure 4: **Text-free proxy benchmark heatmap.** Rows follow the order of the benchmark table: App80 prompted watershed, App80 CPSAM, App80 final multiscale, App65 old multiscale, App65 threshold-hinted Cellpose, App65 final direct-brightfield, App81 threshold-hinted Cellpose, App81 resized multiscale Cellpose, App81 image-generation mask, and App81 final stitched hybrid. Columns show, from left to right: Dice-to-final score, area-ratio agreement, instance-count agreement, low-fragment score, low-edge touch score, and low-dominant-merge score. Teal indicates stronger agreement or lower QC burden; red indicates weaker agreement or higher QC burden.